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Tensor sketch – Scene Sketch Generation Using Structure Tensor

V. B. Surya Prasath
Computational Imaging and VisAnalysis Lab, Department of Computer Science, University of Missouri-Columbia
Columbia, MO 65211, USA
e-mail : surya.iit@gmail.com; Fax/phone : 573-882-8391/573-882-8318

Scene sketch generation is useful for non-photo realistic renderings, and digital entertainment. Recently, computerized sketch generation from digital images for objects, faces, and materials picked up considerable interest and applications are found in law enforcement, image based stylization, and other preliminary tasks such as segmentation that are required to perform pattern recognition. In this work, we utilize multiscale structure tensor based approach that works well in capturing salient boundaries and provides visually pleasing sketch results. Experimental results on two different datasets indicate the usefulness of our proposed approach.

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1 Introduction

Scene sketch generation and texture synthesis are utilized in many computer vision sub-domains such as object recognition, segmentation, face recognition, and materials classification. Due to the ill-posed projective nature of 3D scenes onto 2D domain, the contours of objects contain important visual clues in determining where the objects are and how to identify them correctly. Hence, it is important to have a robust scene sketch generation method that can capture the essential contents in a given scene with robustness to various shape structures that maybe present. In scene sketch literature there are two major directions considered by various researchers in the past. One is the computer-based that depends on exploiting image content to generate both contours and shading texture. The other one are based on approaches that rely on line drawing generation for sketch synthesis. These techniques utilize only shape information and do not, in general, incorporate texture information [1].

Among a wide variety of automatic scene sketch generation methods from the literature we mention the following landmark works. Zeleznik et al [2] studied an interface for combining the advantages of sketching with high-level editing in order to create an environment for rapidly conceptualizing and editing approximate 3D scenes. Wang and Tang [3] considered multiscale markov random fields model for face photo-sketch synthesis and recognition. Recently, Sidike et al [4] proposed a new image scene sketch method based on directional Prewitt differential gradient kernels and Kirsch compass kernels along with relative variance incorporated binary encoding step. Then, the original image pixel intensity values were combined with the output from the encoding step for decoding and obtained scene sketch.

In this work, we propose to use multiscale structure tensor eigenvalues based coherence measure instead of traditional gradient kernels for robustly extracting edges. Structure tensor eigenvalues can be used in detecting image discontinuities robustly unlike traditional gradients based approaches, which are oscillatory under image noise [5]. By using multiscale computations edges of objects with different scales can be captured. We utilize the encoding and decoding steps used in [4] to sketch object and our multiscale structure tensor eigenvalues based edge response. Experiments are conducted on two different datasets, namely the MUCT face database [6], which consist of faces under different lighting, pose angle and ethnicity, and Berkeley segmentation dataset (BSDS) [7], which consist of various natural images. Further, we compare our proposed structure tensor driven method to the gradient edge fusion based method studied in [4].

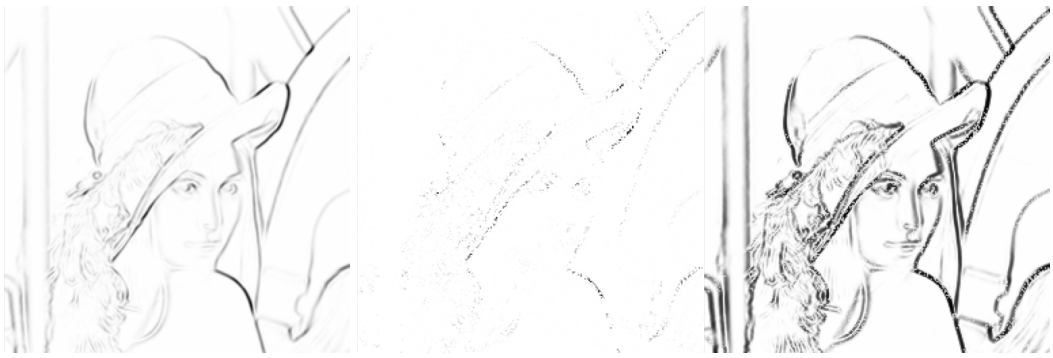

Figure 1. Example of structure tensor eigenvalues and the coherence. Left: λ^{σ}_{+}, Middle: λ^{σ}_{-}, Right: coherence map.

We organized the rest of the paper as follows. Section 2 introduces the structure tensor based scene sketch generation from digital images. Section 3 experimental results on various natural and face image datasets is given with discussion. Finally, Section 4 concludes the paper.

2 Structure Tensor based Scene Sketch Generation

2.1 Edge extraction with multiscle structure tensor

Our main methodology follows the three main stages as in [4], namely, edge extraction, binary encoding, and decoding. However, our proposal differs from the method [4] in that we utilize a robust multiscale structure tensor eigenvalues based edge extraction step that can overcome limitations associated with traditional gradient based edge detectors.

We briefly describe the smoothed structure tensor and its eigenvalues computations here and refer to [5] for further details. Structure tensor, also known as the second moment matrix, of an image I is given as,

$$T^{\sigma}(u,v) = \begin{pmatrix} G_{\sigma} * I_u^2 & G_{\sigma} * I_u I_v \\ G_{\sigma} * I_u I_v & G_{\sigma} * I_v^2 \end{pmatrix} \qquad (1)$$

where (I_u, I_v) is the gradient of the image I and (u,v) pixel location. Here the G_{σ} is the Gaussian kernel with $\sigma \geq 0$ (scale parameter) utilized as a smoothing operation. Let the eigenvalues of this matrix (1) be (pixel-wise dependency omitted for notational clarity) $(\lambda^{\sigma}_{+}, \lambda^{\sigma}_{-})$, where $\lambda^{\sigma}_{+} \geq \lambda^{\sigma}_{-}$ are the maximum and minimum respectively. We form the *coherence* measure as,

$$C^{\sigma}(x,y) = \left(\frac{\lambda_{+}^{\sigma} - \lambda_{-}^{\sigma}}{k + \lambda_{+}^{\sigma} + \lambda_{-}^{\sigma}}\right)^2 \qquad (2)$$

where $k > 0$ is a parameter. There are other options in deriving a coherence measure to capture edges [5, 6] and the above measure (2) is the simplest among them. Figure 1 shows the $(\lambda^{\sigma}_{+}, \lambda^{\sigma}_{-})$ and the coherence map C^{σ} for the standard test Lena image.

2.2 Binary encoding and image decoding

Following [4], a binary encoding with relative local variance as thresholding method is employed to obtain object sketches. We first compute the coherence measure (2) by taking over a range of scales for the eigenvalues as follows.

$$C^{\sigma^k}(x,y) = \left(\frac{\lambda_+^{\sigma^k} - \lambda_-^{\sigma^k}}{k + \lambda_+^{\sigma^k} + \lambda_-^{\sigma^k}}\right)^2$$

Then, the corresponding coherence measure is used in the binary code that is calculated by the following formula,

$$D_k = B(\left|C^{\sigma^k}\right| - RelVar)$$

and

$$B(x) = \begin{cases} 0, & if\ x < 0 \\ 1, & if\ x \geq 0 \end{cases}$$

where *RelVar* is the relative variance computed as,

$$RelVar = \frac{STD(C^{\sigma^k})}{MEAN(C^{\sigma^k})}$$

Here STD is the standard deviation, and MEAN is the mean on the computed coherence map. Finally, the decoding step is performed to get back gray scale sketch image,

$$Q(u,v) = \delta \times \sum_{k=0}^{k=7} (D_k(u,v) \times 2^k)$$

where $Q(u,v)$ is the final decoded pixel value at location (u,v) in the local patch p and δ is a constant.

3 Experimental Results and Discussion

Our method takes on average 4 seconds for a gray scale image of size 256 x 256 using a MATLAB implementation on a 2.3 GHz Intel Core i7, 8GB 1600 MHz DDR3 Mac Laptop. This is twice the amount of computational time taken by the method proposed in [4] due to the computations involved in computing multiscale structure tensor eigenvalues. The number of scales is chosen to be 8 (this is to be used in the binary coding strategy), however further scales can be chosen depending upon the scene's content (in this case the decoding will obtain pixel values higher than 8-bit capacity, a clipping can be used to get the 8-bit [0 255] images). Using multiple scales help capture edge responses better than utilizing a single scale. We chose the scales {1, 2, 3, 4, 5, 6, 7, 8}. There are other sophisticated choices available such as adapting the scales based on the image content locally e.g. local histogram. However, in our tensor sketch approach our aim is to capture the standard range of scales and the chosen scales obtained good sketches with the standard choice. The proposed tensor sketch is not sensitive to the choice of scales as the images (natural and face) are within the 8 scales utilized here. We used two different datasets for sketch generation: MUCT face database [6], consisting of various faces with lighting, pose angle and ethnicity changes, and the Berkeley segmentation dataset (BSDS) [7], consisting of various scenes of people, buildings, and nature.

Figure 2 and 3 show sample outputs from the MUCT and BSDS respectively. We compare our tensor sketch approach with that of gradient kernels based method [4]. As can be seen, our results contain salient contours meanwhile the results from [4] contain major texture content. This is more so in the MUCT face database results (Figure 2) where our tensor sketch approach obtains smooth contours of the major discontinuities whereas the method [4] obtains small scale texture of the hair, moustache etc. Due to the multiscale nature of structure tensor eigenvalues used in our approach, in natural images from BSDS (Figure 3), the salient edges are captured including the transitions of sky to the ground. However, in homogeneous background based images for e.g., birds, our approach obtains coherent sketch whereas the

method [4] obtained oscillatory contours. Perhaps a combination of our approach and the gradient kernels can leverage both the salient contours and texture content respectively and defines one of the future research directions.

Figure 2. Sample outputs of face sketch images on MUCT database [6]. Top row: Original input images. Middle row: scene sketches generated by gradient kernels [4]. Bottom row: our proposed tensor sketch approaches.

Finally, to quantitatively compare the tensor sketch and the gradient kernels method [4] we use the following edge quality comparison metric [8] (strictly speaking this is not an optimal measure for scene sketch generation results, however since both the methods compared here rely on image derivatives this metric is relevant) known as the edge peak signal to noise ratio ($PSNR_E$),

$$PSNR_E(u) = 20 * log_{10}\left(\frac{\max EM\,(I) - \min EM(I)}{MSE_E}\right)$$

where $EM(I) = 2 - (2 /(1+k|\nabla G_\sigma * I|^2))$ is the edge map with $k = 0.0025$, $\sigma = 0.5$, $MSE_E = (mn)^{-1} \sum\sum (EM(u) - EM(I))$. Here we note that the original input image is I, and the scene sketch image is u, and we utilize the canny edge map of the original image I and for $EM(u)$ the final sketch images are taken. We also remark that the Canny edges are not optimal, however we found that for the MUCT and to a large extent for BSDS natural images they provide a general guideline of major edges. Table 1 shows the $PSNR_E$ values for the images shown in Figures 2 and 3. As can be seen, our tensor sketch approach obtained higher $PSNR_E$ values indicating closer affinity towards global edges. The low values for the gradient kernels [4] method are due to the lack of continuous edge contours.

Table 1 Comparison of $PSNR_E$ (dB) values for our proposed tensor sketch and the gradient kernels method [4] on MUCT face database images shown in Figure 2, and BSDS natural images shown in Figure 3.

Image	Gradient kernels [4]	Tensor sketch
Face 1	20.43	23.76
Face 2	21.32	24.40
Face 3	21.87	25.17
Face 4	20.98	24.45
Face 5	20.24	24.18
Butterfly	15.37	18.50
Birds	16.23	19.49
Mountains	15.92	19.18
Deer	15.76	19.22

4 Conclusion

This work presented an automatic scene sketch generation from digital images utilizing multiscale structure tensor based edge coherence. By computing multiscale structure tensor eigenvalues and using a smooth coherence map we obtained better salient edges and regions than traditional edge detectors based approach. Experimental results on two different datasets showed that our proposed tensor sketch method is able to retain better image features without compromising visually pleasing scene sketches generation. Extension to color imagery is natural due to the involvement of structure tensors and further directions include adding texture specific features maps for better capture of small-scale details, and fusing with gradient kernels.

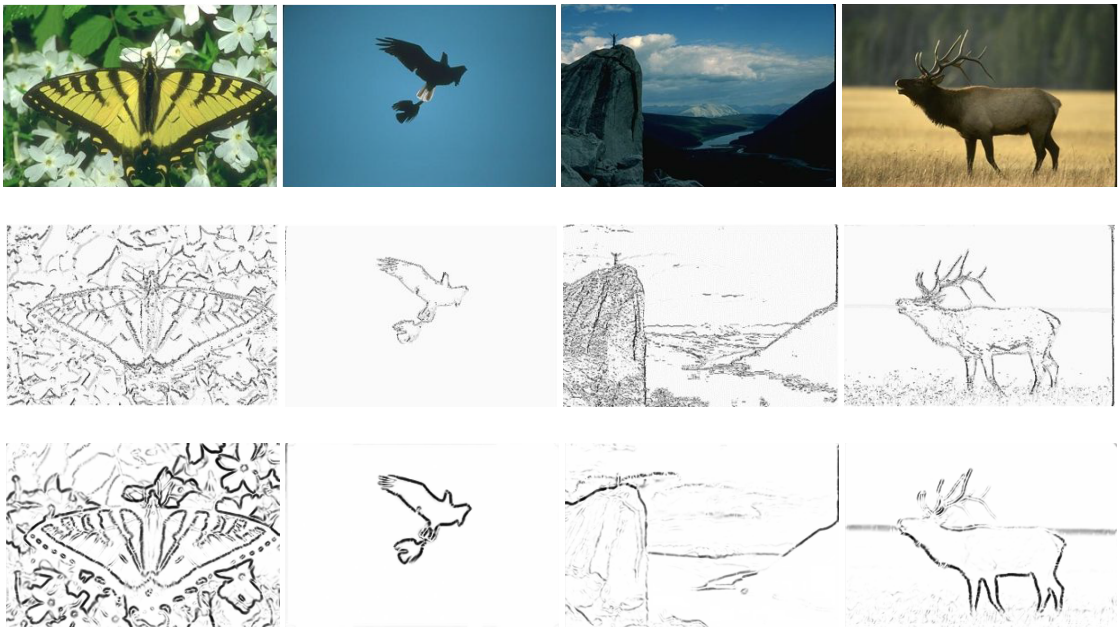

Figure 3. Sample outputs of face sketch images on BSDS database [7]. Top row: Original input images. Middle row: scene sketches generated by gradient kernels [4]. Bottom row: our proposed tensor sketch approaches.

4 References

1 Freeman W, Tenenbaum J, Pasztor E, ACM Trans Graph, 22(2003) 33

2 Zeleznik RC, Herndon KP, Hughes JF, ACM SIGGRAPH courses (2007)

3 Wang X, Tang X, IEEE Trans Patt Anal Mach Intel, 31(2009) 1955

4 Sidike P, Essa A, Asari V, Opt Patt Recog XXVII, Proc SPIE, 9845(2016)

5. Prasath VBS, Informatica, 40(2016) 337

6 Milborrow S, Morkel J, Nicolls F, Patt Recog Assoc South Africa, (2010)

7 Martin D, Fowlkes C, Tal D, Malik J, Int Conf Comp Vis, 2(2001) 416

8 Guo Z, Sun J, Zhang D, Wu, B, IEEE Trans Image Proc, 21(2012) 958